📄 DOCUMENT 7

Physical Infrastructure & MESOB Building Setup Blueprint

Hawassa MESOB One-Stop Service Center

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PART 1

(Sections 1–5: Facility Design Philosophy, Spatial Zoning Plan, Service Window Modeling, Capacity & Throughput Planning, Core Physical Infrastructure Requirements)

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1. FACILITY DESIGN PHILOSOPHY

The MESOB One-Stop Service Center must not resemble a traditional government office.

It must be:

- Efficient
- Transparent
- Digitally enabled
- Citizen-centric
- Performance-driven
- Scalable for future expansion

The building must reflect reform.

Citizens should immediately experience:

- Order
- Clarity
- Reduced confusion
- Professional service environment

1.1 Design Objectives

The building must:

1. Support 47 Phase 1 services
2. Accommodate 1,000–2,500 daily transactions
3. Allow 50–70 concurrent service interactions
4. Provide space for technical infrastructure
5. Enable secure operations
6. Support Phase 2 expansion without major redesign

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2. SPATIAL ZONING PLAN

The building must be divided into functional zones.

2.1 Zone A – Public Reception & Queue Area

Functions:

- Entry screening
- Service selection kiosk
- Ticket issuance
- Waiting area
- Information desk

Requirements:

- Digital queue display screens
- Seating for minimum 150–250 people
- Clear service categorization boards
- Public announcement system

2.2 Zone B – Service Counter Hall

Primary operational zone.

Must include:

- 25–40 service counters (Phase 1 target: 30)
- Each counter equipped with:
    - Desktop PC
    - Dual monitors
    - Receipt printer
    - POS terminal
    - Document scanner
    - Secure storage drawer

Counters grouped by service domain:

- Land counters (8–10)
- Revenue counters (6–8)
- Trade & Investment counters (6–8)
- Health/Education counters (3–4)
- General inquiry & complaint counters (2–3)

2.3 Zone C – Bureau Review Offices

Back-office review rooms for:

- Land officers
- Revenue validation staff
- Technical review teams

Purpose:

- Digital approval processing
- Document verification
- Escalation resolution

Minimum 6–10 offices required.

2.4 Zone D – Management & Executive Area

Includes:

- MESOB Director office
- SLA Monitoring Unit office
- Technical support office
- Meeting room
- Executive review room

2.5 Zone E – Secure Infrastructure Room

Restricted access.

Must contain:

- Network core switch
- Firewall
- Server racks (if local hardware)
- UPS system
- Backup appliance

Security requirements:

- Biometric access
- CCTV monitoring
- Environmental control

2.6 Zone F – Document Archival Room

For:

- Temporary storage of scanned legacy records
- Physical backup files (if required)

Phase 1 goal:

Minimize physical storage dependency.

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3. SERVICE WINDOW MODELING

Service windows must be optimized by transaction type.

3.1 Counter Time Estimation

Average processing time per service:

Service Category	Avg Processing Time
Revenue Payment	5–10 minutes
License Renewal	10–15 minutes
Lease Transfer	20–30 minutes
Investment Registration	20–40 minutes

Weighted average across services:

~15 minutes per transaction

3.2 Throughput Calculation

If:

- 30 counters operational
- Each handles 4 transactions per hour
- 8-hour service day

Daily capacity:

30 × 4 × 8 = 960 transactions/day

With peak optimization & quick services:

Can scale to 1,200–1,500/day

Matches projected Phase 1 load.

3.3 Counter Distribution Strategy

Counters must be:

Flexible and reassignable.

System must allow:

- Dynamic service reallocation
- Peak-hour balancing
- Real-time performance monitoring

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4. CAPACITY & MANPOWER PLANNING

4.1 Required Staffing (Phase 1)

Counter Staff

- 30 service officers
- 5 floating backup officers

Bureau Review Officers

- Land: 4–6
- Revenue: 3–4
- Trade/Investment: 3–4
- Health/Education: 2–3

Technical Staff

- 1 IT infrastructure manager
- 2 IT support officers
- 1 integration coordinator

Management

- MESOB Director
- SLA Monitoring Officer
- Finance reconciliation officer

Estimated Total Staff:

55–70 personnel

4.2 Shift Model (Optional)

If demand increases:

Introduce staggered shifts:

- Morning shift
- Afternoon shift

Building must allow 12-hour extended operation if needed.

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5. CORE PHYSICAL INFRASTRUCTURE REQUIREMENTS

5.1 Electrical Infrastructure

Must include:

- Dedicated power distribution panel
- UPS coverage for:
    - All counters
    - Network core
    - Servers
- Generator backup
- Surge protection

5.2 Structured Cabling

- CAT6 or higher cabling
- Central patch panel
- Labelled network ports per counter
- Separate VLAN for security zones

5.3 Environmental Controls

- Air conditioning for counter hall
- Climate control for server room
- Fire detection & suppression
- Emergency lighting

5.4 Accessibility Requirements

Building must include:

- Disability-accessible entry
- Accessible counters
- Clear signage
- Crowd flow management design

5.5 Security Infrastructure

- CCTV coverage
- Access control system
- Visitor logging
- Security personnel desk

PHYSICAL CAPACITY SUMMARY (PART 1)

The MESOB building must support:

✓ 30+ counters
✓ 1,000–1,500 daily transactions
✓ 55–70 staff
✓ Secure infrastructure room
✓ Structured network & power redundancy
✓ Public comfort & accessibility

This physical blueprint aligns with Phase 1 digital architecture.

END OF DOCUMENT 7 – PART 1

Covered:

✓ Facility design philosophy
✓ Spatial zoning model
✓ Service window modeling
✓ Capacity planning
✓ Core infrastructure requirements

Reply with:

“Proceed with Document 7, Part 2.”

Next we will produce:

✓ Detailed hardware placement plan
✓ Network topology inside building
✓ Queue system deployment plan
✓ Expansion readiness for Phase 2
✓ Cost drivers & renovation considerations.

📄 DOCUMENT 7

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PART 2
(Sections 6–10: Hardware Placement Plan, Internal Network Topology, Queue System Deployment, Phase 2 Expansion Readiness, Renovation Cost Drivers)

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6. DETAILED HARDWARE PLACEMENT PLAN

The hardware layout must directly support the enterprise architecture defined in Document 6.

Physical placement must ensure:

- Operational efficiency
- Security segmentation
- Maintainability
- Future scalability

6.1 Service Counter Hardware Layout

Each of the 30 Phase 1 counters must include:

1. Desktop workstation (minimum i5 equivalent or higher)
2. Dual monitor setup (one internal, one citizen-facing optional)
3. Thermal receipt printer
4. POS terminal (integrated with payment gateway)
5. Document scanner (flatbed or compact high-speed)
6. Token display indicator (linked to queue system)
7. Secure drawer for cash handling (where applicable)

Power & Network:

- Dedicated network port (VLAN-segmented)
- Dedicated power socket (UPS-backed)
- Cable management concealed for security

6.2 Bureau Review Office Equipment

Each review officer desk must include:

- Desktop or laptop workstation
- Dual monitors (for document review)
- Secure network port
- Access to document management system

Shared resources:

- High-speed network printer
- Large-format scanner (for land maps)
- Secure file storage cabinet (temporary only)

6.3 Infrastructure Room Hardware Placement

Server Rack Layout (if on-prem equipment used):

Top Rack:

- Core switch
- Firewall appliance
- Router

Middle Rack:

- Application server(s)
- Database server(s)
- Storage appliance

Lower Rack:

- Backup appliance
- Patch panel
- Power distribution unit (PDU)

UPS units must be floor-mounted with ventilation clearance.

6.4 CCTV & Security Equipment Placement

Cameras must cover:

- Entrance
- Waiting area
- All service counters
- Infrastructure room entrance
- Cash handling areas

All camera feeds must route to:

- Secure monitoring console
- Storage system with minimum 30-day retention

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7. INTERNAL NETWORK TOPOLOGY

The internal building network must mirror the logical security zoning.

7.1 Core Network Architecture

Internet/Fiber Entry
↓
Edge Router
↓
Firewall
↓
Core Switch
↓
Access Switches (per zone)

7.2 VLAN Segmentation Model

Minimum VLAN separation:

- VLAN 1: Service Counters
- VLAN 2: Bureau Review Offices
- VLAN 3: Server/Infrastructure
- VLAN 4: Management & Admin
- VLAN 5: CCTV
- VLAN 6: Guest Wi-Fi (if provided)

No cross-zone traffic without firewall policy control.

7.3 Redundancy Requirements

- Dual uplinks from core switch
- Backup ISP link
- Automatic failover configuration
- Redundant power supply for core switch

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8. QUEUE SYSTEM DEPLOYMENT PLAN

Queue system must integrate with workflow engine.

It must not operate independently.

8.1 Queue Components

- Ticket issuance kiosk (2–4 units)
- Digital display screens (5–10 across hall)
- Counter display indicators
- Central queue server (or cloud module)
- SMS notification module (optional)

8.2 Queue-to-Service Integration

When citizen selects service:

1. Ticket generated
2. Service category logged

3. Workflow instance pre-created
4. Counter assignment triggered
5. SLA timer initiated upon service start

Queue system must feed analytics dashboard.

8.3 Performance Metrics Captured

Queue system must capture:

- Average waiting time
- Service time per counter
- Peak-hour distribution
- Counter utilization rate

These feed into:

- SLA monitoring dashboard
- Monthly executive review

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9. PHASE 2 EXPANSION READINESS

The building must allow expansion without major structural redesign.

9.1 Counter Expansion Capacity

Design hall to allow:

- Additional 10 counters (Phase 2)
- Movable partitions
- Modular furniture configuration

9.2 Infrastructure Scalability

Server room must support:

- Additional rack space
- 50% additional cooling capacity
- Expanded UPS load
- Additional fiber ports

9.3 Space Allocation Buffer

Reserve:

- 1–2 rooms for future GIS unit
- 1 additional meeting room
- Storage for digitization equipment

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10. RENOVATION COST DRIVERS

Major cost categories include:

1. Civil works (partitioning, flooring, painting)
2. Electrical upgrades
3. Structured cabling installation
4. HVAC installation
5. Security system installation
6. Furniture & counter construction
7. Server room construction
8. Accessibility modifications

Cost variables depend on:

- Existing building condition
- Square meter size
- Electrical capacity of current structure

10.1 Renovation Risk Factors

- Underestimated electrical upgrade need
- Insufficient cooling for server room
- Overcrowding risk due to underestimated traffic
- Poor acoustic planning causing noise issues

Mitigation:

Pre-renovation technical assessment required.

PHYSICAL INFRASTRUCTURE SUMMARY

The MESOB building must be:

✓ Digitally integrated
✓ Securely segmented
✓ Capacity-aligned with transaction projections
✓ Scalable for Phase 2
✓ Professionally designed for citizen experience

The physical environment must reflect the seriousness of governance reform.

DOCUMENT 7 COMPLETE

This concludes:

✓ Hardware placement plan
✓ Internal network topology
✓ Queue system deployment
✓ Expansion readiness planning
✓ Renovation cost drivers

Reply with:

"Proceed with Document 8, Part 1."

Next we begin:

Comprehensive Procurement & Bill of Quantities (BoQ)
(Strictly aligned to Ethio Telecom Sidama region item list and mapped to Hawassa MESOB acquisition demand).